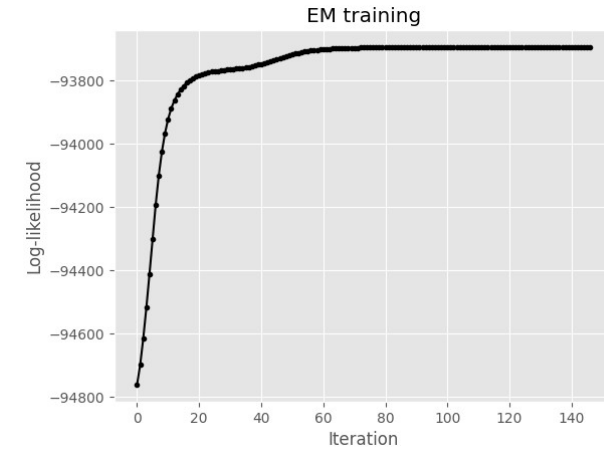
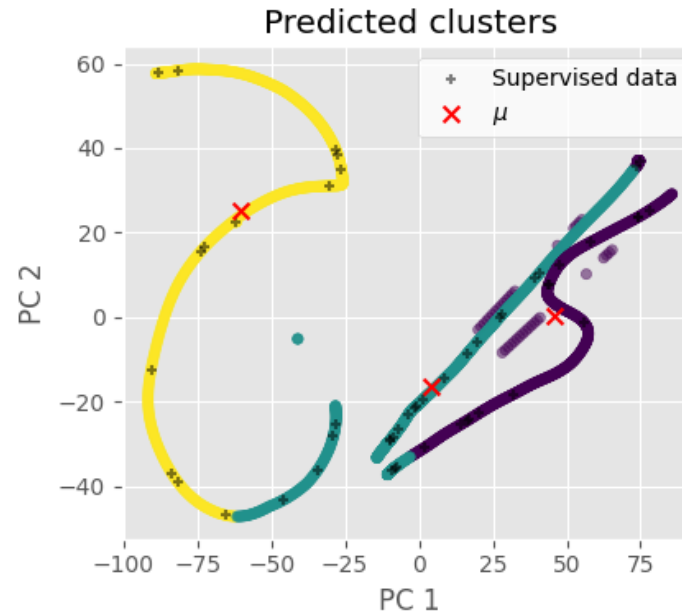
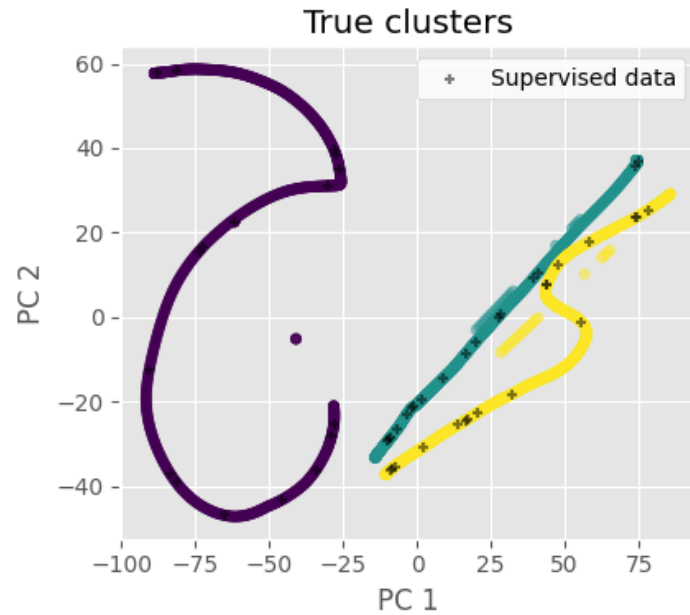


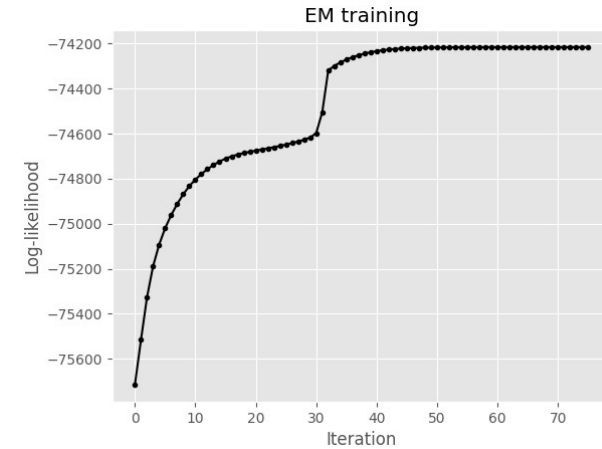
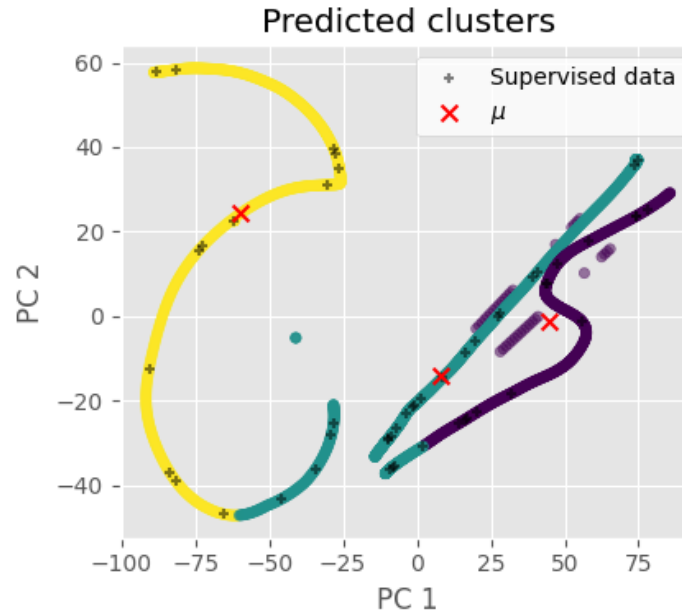
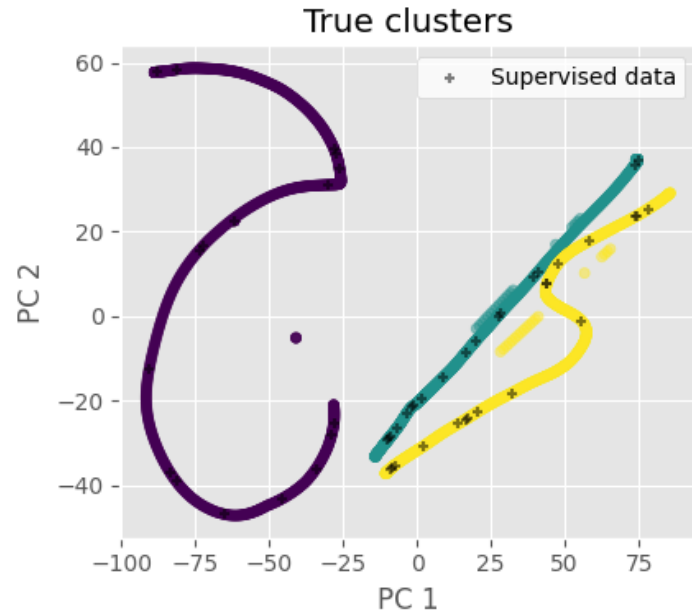
Semi-supervised study
2026/06/24

Fully **u**nsupervised ($\beta = 0$, full covariance)



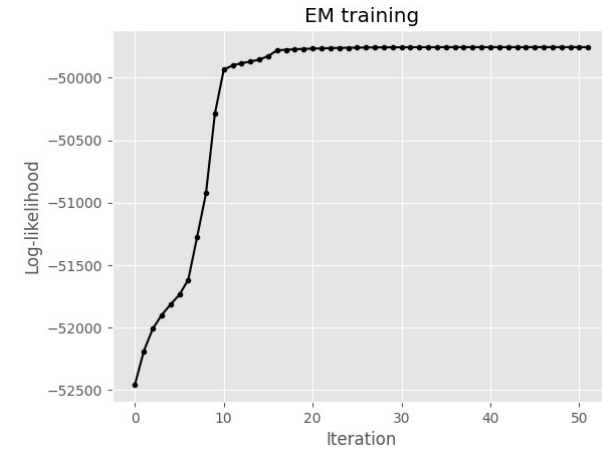
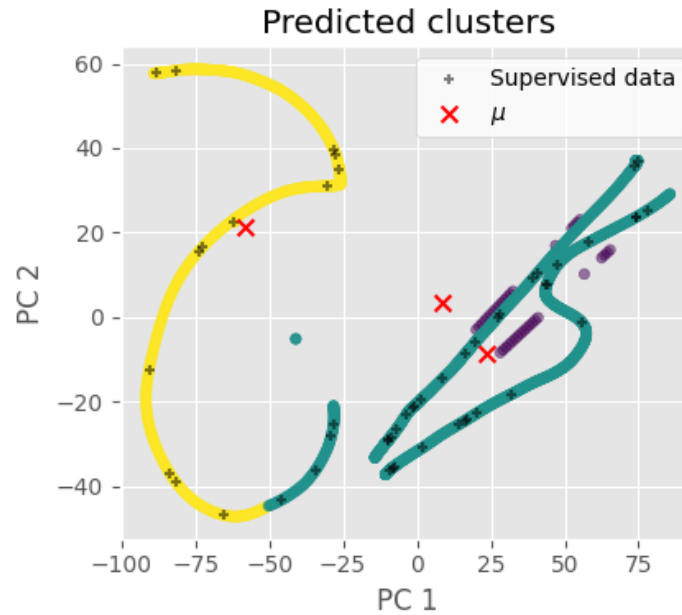
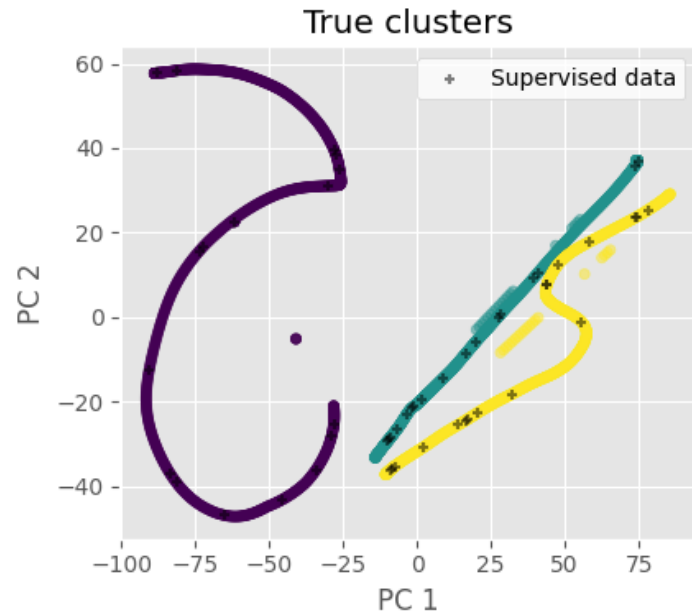
- Silhouette: 0.2998111357081737
- Completeness: 0.6273980349564303
- Fowlkes-Mallows index: 0.7451652521812167
- Homogeneity: 0.6176323837102664
- Mutual Information: 0.678538526623476
- Adjusted mutual information: 0.622251095013317
- Normalized Mutual Information: 0.6224769098898842
- Rand index: 0.827184794922935
- Adjusted rand index: 0.6144056423445312
- V-measure: 0.6224769098898842

Mostly **unsupervised** ($\beta = 0.25$, full covariance)



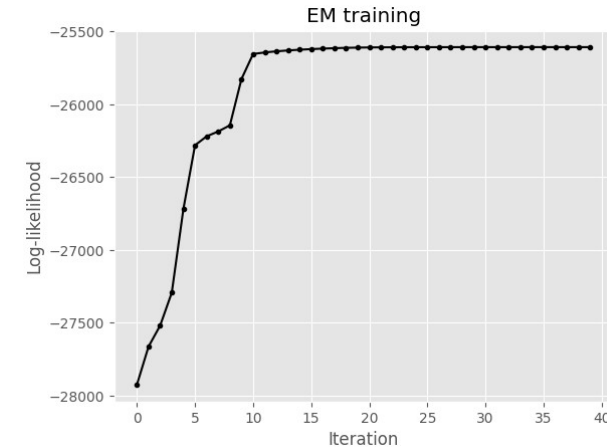
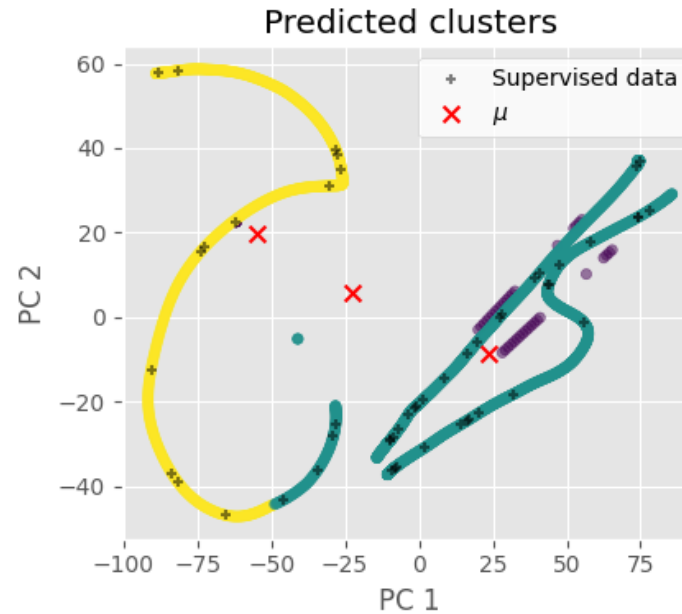
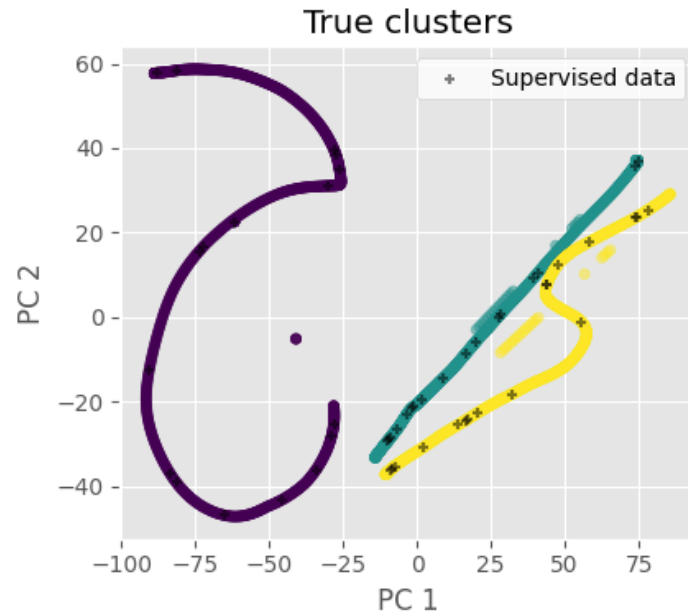
- Silhouette: 0.2842655695746486
- Completeness: 0.6725440846120307
- Fowlkes-Mallows index: 0.7632944195196653
- Homogeneity: 0.6544388750330659
- Mutual Information: 0.7189745902934597
- Adjusted mutual information: 0.6631654472441469
- Normalized Mutual Information: 0.6633679670782755
- Rand index: 0.8371747123629654
- Adjusted rand index: 0.6391212599806535
- V-measure: 0.6633679670782754

Balanced objective ($\beta = 0.5$, full covariance)



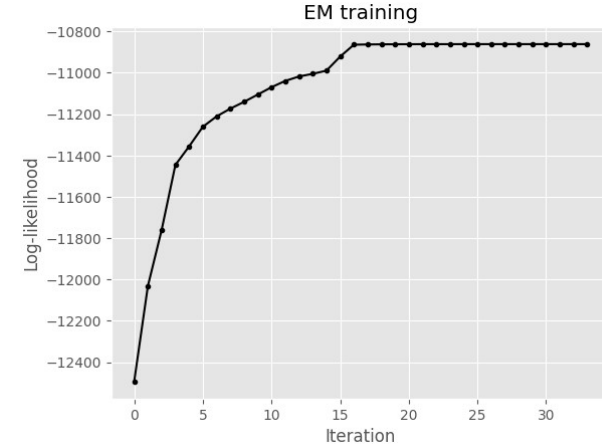
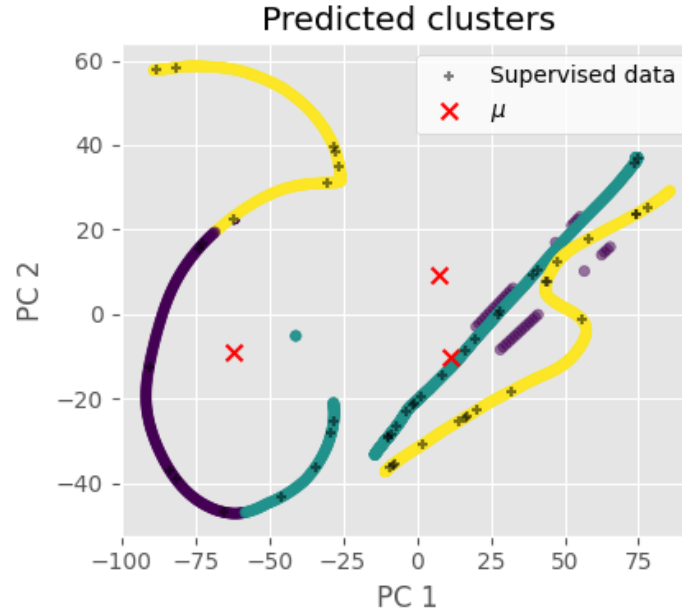
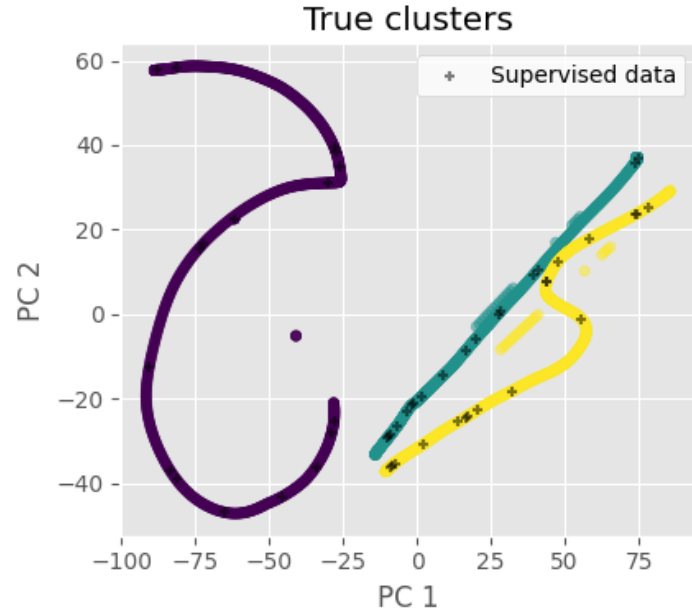
- Silhouette: 0.34894549310751355
- Completeness: 0.6891718971298522
- Fowlkes-Mallows index: 0.6753782353415967
- Homogeneity: 0.41248162633189894
- Mutual Information: 0.45315738353803164
- Adjusted mutual information: 0.5157175437276091
- Normalized Mutual Information: 0.5160801266392561
- Rand index: 0.6819488681075654
- Adjusted rand index: 0.3950955506838093
- V-measure: 0.5160801266392561

Mostly supervised ($\beta = 0.75$, full covariance)



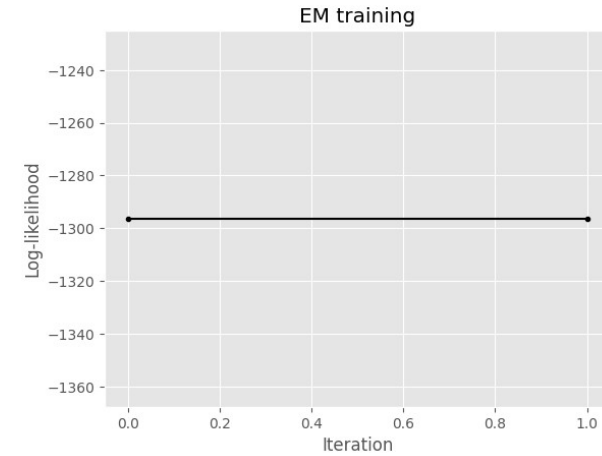
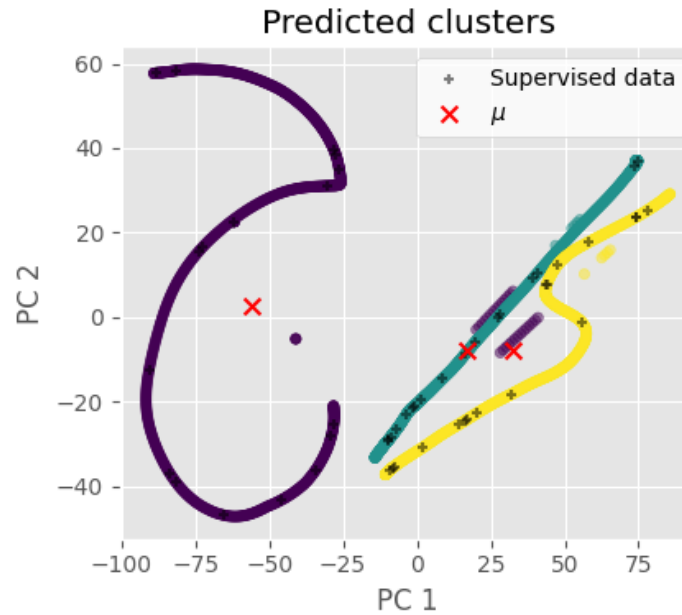
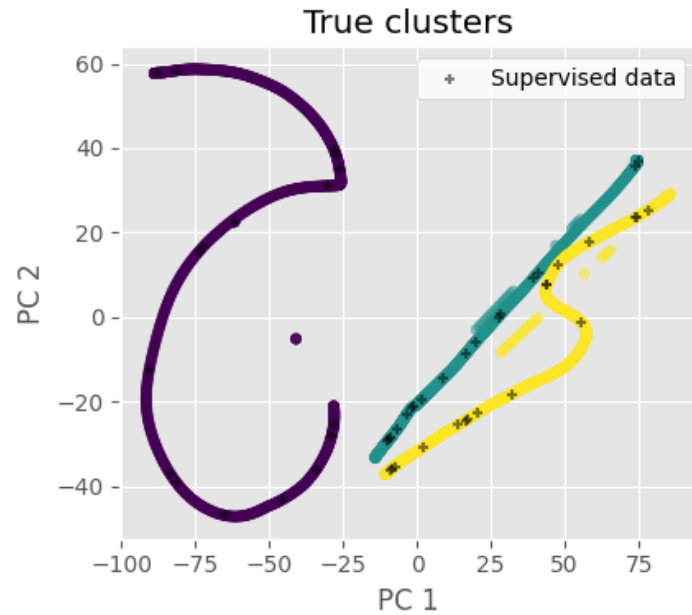
- Silhouette: 0.4088628142181719
- Completeness: 0.6640332808802846
- Fowlkes-Mallows index: 0.6732149630110269
- Homogeneity: 0.4070363470130773
- Mutual Information: 0.44717513276314386
- Adjusted mutual information: 0.5043355872848532
- Normalized Mutual Information: 0.5047023534337957
- Rand index: 0.6819912674346033
- Adjusted rand index: 0.3936453765321247
- V-measure: 0.5047023534337957

Mostly supervised ($\beta = 0.9$, full covariance)



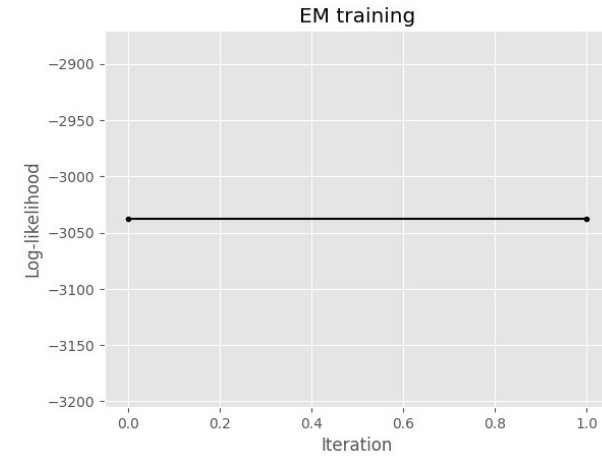
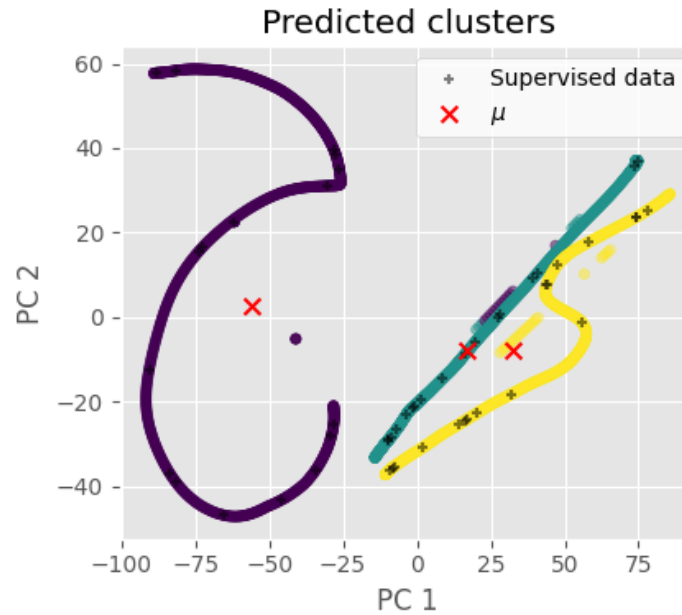
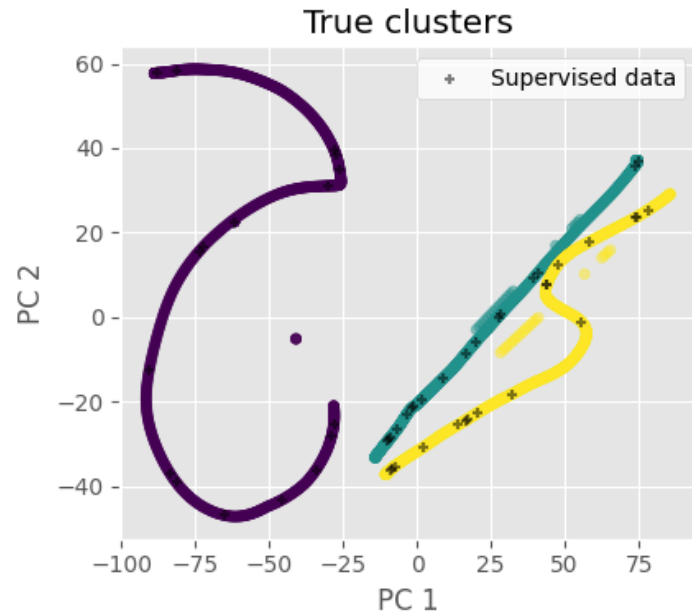
- Silhouette: 0.09112858146983326
- Completeness: 0.5839133377632219
- Fowlkes-Mallows index: 0.6919092045630955
- Homogeneity: 0.5017036186283245
- Mutual Information: 0.5511777606943362
- Adjusted mutual information: 0.5394017121321106
- Normalized Mutual Information: 0.5396957606389736
- Rand index: 0.7642313341202649
- Adjusted rand index: 0.5020994559197169
- V-measure: 0.5396957606389735

Fully supervised ($\beta = 1$, full covariance)



- Silhouette: 0.21096108419347445
- Completeness: 0.9561412864968315
- Fowlkes-Mallows index: 0.981818800576102
- Homogeneity: 0.9559793330209151
- Mutual Information: 1.0502506429695209
- Adjusted mutual information: 0.9560342237481432
- Normalized Mutual Information: 0.9560603029002765
- Rand index: 0.9878848162921958
- Adjusted rand index: 0.9727345048839638
- V-measure: 0.9560603029002765

Fully supervised ($\beta = 1$, **diagonal** covariance)



- Silhouette: 0.21526276215085657
- Completeness: 0.9806541826971374
- Fowlkes-Mallows index: 0.9922260415149939
- Homogeneity: 0.9806133204075569
- Mutual Information: 1.0773138442313808
- Adjusted mutual information: 0.9806222575803965
- Normalized Mutual Information: 0.9806337511266717
- Rand index: 0.9948204982090524
- Adjusted rand index: 0.9883425897841498
- V-measure: 0.9806337511266717